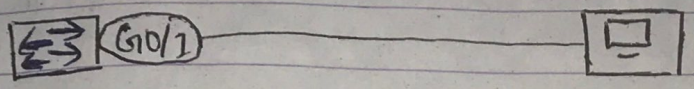
Day-21

Nov-22

PortFast (STP Toolkit) Part-I.

Port-Fast the problem.



When an end host connects to a switch port, the port becomes up/up but can't send/receive data yet.

- It is is a designated port but will take 30 secs before it enters the forwarding state.
  - 15 secs in ~~learn~~ listening
  - 15 secs in learning.

- This leads to a poor user experience.
  - The user probably doesn't even know STP exists.
  - They just know "the interface doesn't work" for 30secs when they connect their computer.

  - This wait is unnecessary because there is no risks of a layer 2 loop occuring between a switch/PC.

Port-Fast the solution.

When PortFast is configured on a port, the port immediately enters the Forwarding state when connected to another device.

- It bypasses Listening/learning and can send/receive data right away.

- You can configure Portfast in two ways:

1).

1) Interface config mode:
SW1 (config-if) # spanning-tree portfast.
This enables Portfast only on the individual interface.

2) Global config mode.
SW1(config)# spanning-tree portfast default.
This enables portfast on all access ports.

- Connection between switchports are almost always trunk links.
- Connections to end hosts are almost always access links

Even if we configure spanning-tree portfast on a trunk port, it won't be active.

- show spanning-tree interface interface-name detail.

- There are 2 kinds of Portfast.
  - edge
  - network.

- edge is the kind we'll be covering for CCNA ~~in this vid~~.
- network is used for feature called Bridge Assurance (not a CCNA topic)

- To disable. Portfast on specific access port
- SW1 (config-if) # spanning-tree portfast disable.

- show spanning-tree interface g0/2 detail.

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Portfast on trunk ports

The std. PortFast configuration commands only enable Portfast on access ports.

- Per-Port (config-if)# spanning-tree portfast.
- Default (SW1 (config-t) # spanning-tree portfast default.

On some cases, we migh want to enable Portfast on a trunk port.

- A port connected to a virtualization server with vitual machines (VMs) in ~~dff~~ different VLANS.
- A port connected to a router via router-on-a stick (ROAS)

This can only be configured per port on in # interface config mode.

SW1 (config-if)# spanning tree portfast trunk.

PortFast edge.

On modern CISCO switches, if we use the commands mentioned earlier, the device will automatically add the edge keyword to configuration.

spanning-tree portfast
Spanning-tree portfast edge.

- Spanning-tree portfast disable doesn't use edge keyword

show running-interface int_name to view the running-configs for specified interface.

Day-4 21 (Part) STP ToolKIT (Part-II) Dec-2

BPDU Guard & BPDU Filter

Portfast & BPDUs

- Portfast makes a port start in the forwarding state when it is connected, but it doesn't disable STP on the port.
  - The port will continue to send BPUs every 2 seconds
- Because end hosts don't run STP and send BPDUs, a ~~portfast~~ Portfast-enabled port shouldn't receive BPDUs.
- If a portfast enabled port receieves an STP BPDU, it will revert to acting like a regular STP port (without Portfast)

BPDU Guard: the problem.

Portfast should only be enable on ports connected to non-switch devices (end hosts, routers).
- These devices don't send BPDUs.

- A portfast-enabled port still send BPDUs and will operate like a regular STP port if it receives BPDUs from a neighbour.
- If an end user cardessly connects a switch to a port meant for end hosts, it could affect STP topology.
  - BPDU Guard acts as a safeguard against this

BPDU Guard the solution.

- BPDU Guard Protects the network from unauthorized switches being connected to ports intended for end
- hosts

- It can be configured separately from PortFast, but both features are usually used together.
  - They both enhance STP's functionality on port intended for end hosts

- A BPDU guard-enabled port continues to send BPDUs, but if it receives a BPDU it enters the error-disabled state.
  - In effect, this disables the port

BPDU Guard configuration.

Like PortFast, BPDU Guard can be configured in two ways

Per Port Port: SW1 # con int g0/1
SW1 # spanning-tree bpduguard enable.

Default. SW1 # spanning-tree portfast bpduguard default.
- When enabled by default, BPDU Guard is activated on all Portfast-enabled ports.

- Use "spanning-tree bpduguard disable" in interface config mode to disable it on specific ports

ErrDisable

It is a CISCO switch feature that disables a port under certain conditions, such as BPDU Guard Violation.
- Some other causes to disable a ports are:-
  - Power Policing violations
  - Port Security violations
  - DAI (Dynamic Arp Inspections) violations

To reenable an error-disabled port, first solve the underlying issue
! If we re-enable the port without fixing the issue, it will just be error disabled again.

We
You can re-enable an err-disabled port in two ways:
Manual: use "shutdown" & "no shutdown" to reset the disabled port.
Automatic: ErrDisable Recovery

ErrDisable Recovery
It is a feature that automatically re-enables err-disabled ports after a certain period of time.

show errdisable recovery
Errdisable recovery is disabled by default.

- The default recovery timer is 300 seconds (5 minutes)
- Err-disabled interfaces will be automatically re-enabled after 5 minutes.

- Use "err-disa" errdisable recovery interval {seconds}" to in global config mode to modify the interval

- Use "errd errdisable recovery cause {cause}" to enable ErrDisable Recovery for ports disabled by a particular cause

BPDU Filter - The problem

A switchport connected to an end host continues sending BPDU every 2 seconds.
- regardless of whether PortFast and/or BPDU Guard are enabled.

- If the port doesn't connect to a switch, sending BPDUs is unnecessary and undesirable for a couple of reasons.
  - Sending BPDU uses some bandwidth & processing power of switch (though very small).
  - It contains LAN's STP topology. So, if max-security is a concern, we should avoid sending this info to users.

- BPDU Filter. solves this problem by preventing a port from sending BPDUs

BPDU Filter: The solution

BPDU Filter stops a port from sending BPDUs.
Unlike BPDU Guard, it doesn't disable the port if it receives BPDU.

- BPDU Guard Filter can be enabled in two ways:
- Per-port: # Spanning-tree bpdufilter enable
  - The port won't send BPDUs
  - The port will ignore any BPDUs it receives with
  - In effect, this disables STP on the port. Use cautiously

Default: spanning-tree portfast bpdufilter, default

- It will enable bpdufilter on all Port-fast enabled ports
- We can use "spanning-tree bpdufilter disable" to disable it on specific ports

bpdufilter enabled port →

- The port won't send BPDUs
- If the port receives BPDU, PortFast & BPDU filter are disabled, and it operates as a normal STP port.

Recommendations:

Enable PortFast and BPDU Guard however we prefer (per-port or default).

- Only enable BPDU filter by default (global config mode) unless we've a very good reason to enable it per-port

BPDU guard & BPDU filter can be enabled on the same port at same time:

i) If BPDU Filter is enabled in global config mode & port receives a BPDU
   a) BPDU filter will be disabled.
   b) BPDU Guard will be trigered (and err-disable the int)

ii) If BPDU filter is enabled on int-config mode & it receives a BPDU.
   i) the BPDU will be ignored
   ii) BPDU Guard won't be trigered

Day-21 RSA PART-III Dec-12.

Root Guard (STP Toolkit).

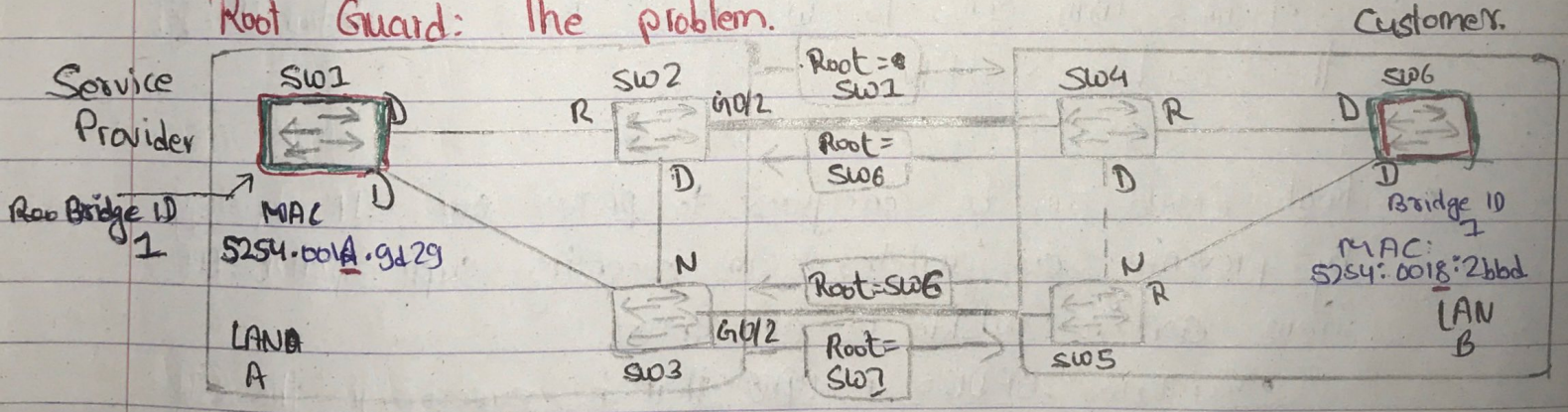
Root Bridge Placement.

STP prevents loop by electing a root bridge and ensuring that each other switch has only one valid path to reach it (root bridge)

- We shouldn't ~~elect~~ randomly select the root bridge. We should consider following things while collecting:-
  - Optimal traffic flow
    - minimize latency
    - minimize congestion.
  - Stability and reliability

We want our root switch to remain operation for as long as possible. The more reliable & more advanced switch shall be used to made root bridge

Root Guard: The problem.



(The fig changes as per condition)

Within our LAN, we can easily control the root bridge by setting its priority to 0. But there are cases where we might connect our LAN to other switches outside of our direct control:-

For eg: ISP provide Metro Ethernet services to customer. It (Metro Eth. Serv) is often used to connect sites within

a MAN.

- In this case even if we set root bridge's priority to 0, its role can be taken by another switch with lower MAC address.

  For eg: In our LAN A root bridge was SW1 with bridge priority 1, SW1, SW2 & SW3 were in our control. But we connected to customer's LAN (LAN B), where switch 6 was root bridge with bridge ID of 1 but lower MAC address. So, the root bridge would shift to SW6 (LAN B), which ain't in our control. So, we don't want this :)

- With no safeguard in place, SW1, SW2 & SW3 accepts SW6 as the root bridge, affecting the service provider's STP topology.
  - Frames from SW3 to SW1 must take a detour through the customer's LAN.

- Root Guard can be configured to protect our STP topology by preventing our switches from accepting ~~supp~~ superior BPDUs from switches outside of our control.
  - Superior BPDU = a BPDU that is superior in the STP algorithm (eg. claiming a better root bridge ID).

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If we want to ensure that root bridge remains in our LAN, we can configure a Root Guard on the ports connected to switches outside of our control.

- SW2 G0/2, SW3 G0/2. (in our case /eg)

- To enable root guard on a port use the below command.
  SW2 (config-if) # spanning-tree guard root

- If a root Guard-enabled port receives a BPDU, it will enter the Broken (Root Inconsistent) state, effectively disabling it.
  - The port won't be able to forward data frames and will discard any frames it receives.
  - SW1, SW2 and SW3 won't accept SW6 as root bridge (Even though LAN A & B are connected they can't communicate now)

- SW-2 G0/2 & SW3 G0/2 as well as ports they connect to (LAN B) on SW4 & SW5 are all designated ports. Normally there can only be one designated port per link, but in this case the switches disagree on ~~which~~ who the root bridge is.

- To re-enable a port disabled by Root Guard, you must solve the issue that disabled the port.
  - Tell the disabled port must stop receiving superior BPDUs.
  - Tell the customer to increase the priority value of their switch

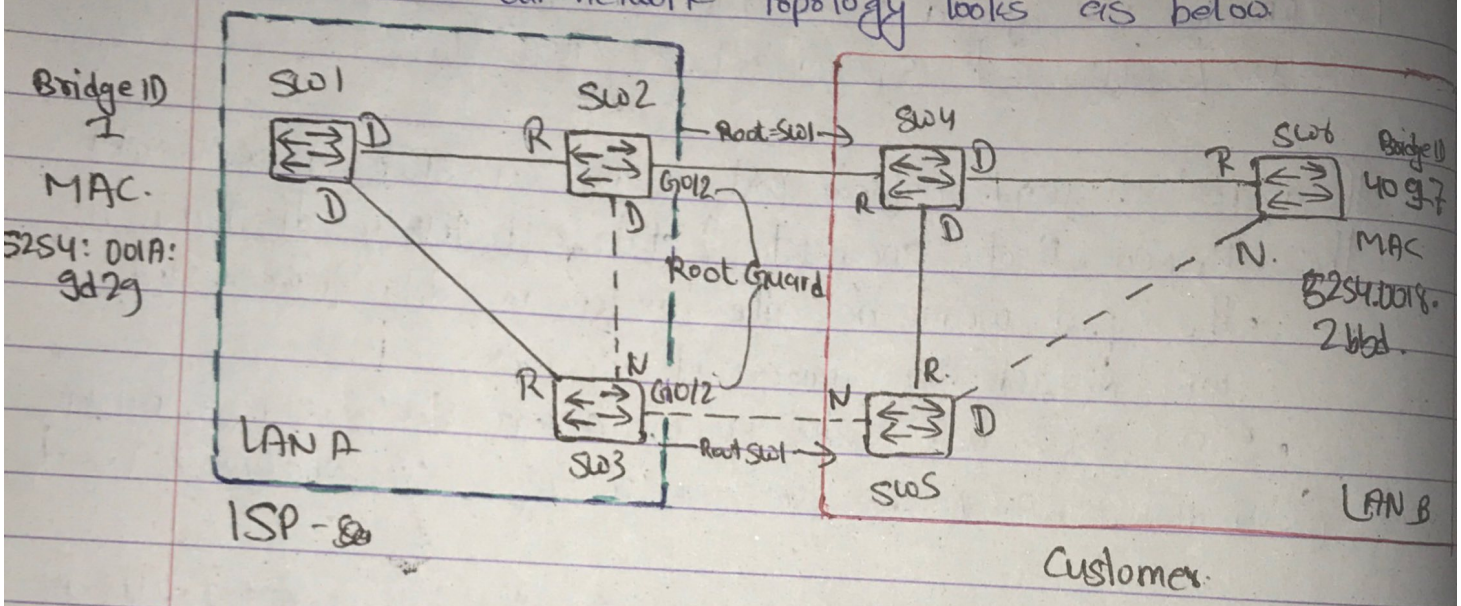
- Once the superior BPDUs received by SW2 G0/2 and SW3 G0/3 age out, the port will automatically be

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re-enabled.

- A BPDU's max Age is 20 sec by default.

After root-guard has been enabled & all problems have been solved our network topology looks as below

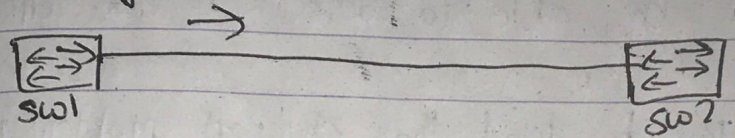


Day-21 (Part-IV). Dec-12.

Root Guard

Unidirectional & Bidirectional link.

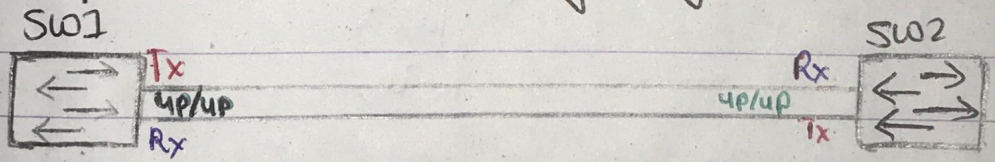
A unidirectional link is a network link where data transmission occurs in only one direction.



eg SW1 can send frames to SW2, but SW2 can't send frames to SW1.

- Unidirectional links are typically caused by layer 1 (Physical) issues.
  - Damaged cables.
  - Faulty connectors or transceivers (eg. SFP modules)

- Unidirectional links are more common with fiber optic cables than the copper UTP.
  - Fibre optics ~~cable~~ connections typically uses two separate fibers.



Although they are separate fibres, these two fibres are considered as one cable, connecting one int on SW1 to one int on SW2.

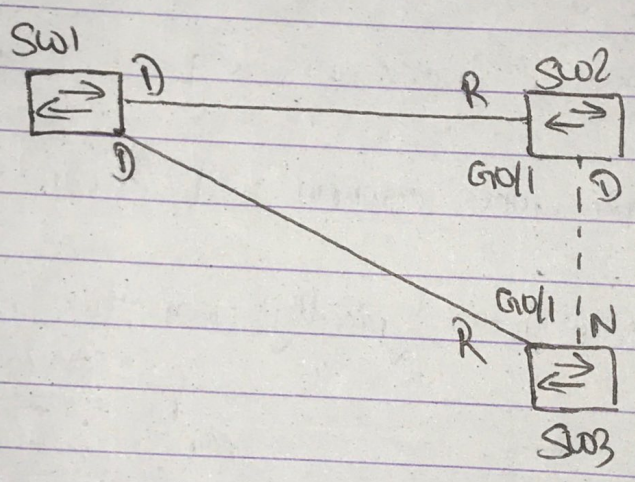
- If one fibre is damaged, it can disrupt data flow in one direction while the other remains unaffected.

- Fibre optics are more vulnerable to damage than copper UTP cables.

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- For a fibre optic cable to be up/up, both ~~the~~ fibres must be connected and functional.
- If there is a physical problem with either fiber, the devices should be able to detect it and disable their interfaces.
- If the devices fail to detect the physical problem, it could result in unidirectional link.

Loop Guard: the problem.



Loop Guard doesn't prevent unidirectional links, but it can protect against the -ve effect of a undirectional link has on STP.

- BPDUs originate from the Root Bridge and are forwarded out of Designated ports.
- SW3 Gi0/1 is a Non-designated port because it receives Superior BPDUs from SW2.
  i.e SW2 has superior root cost & or bridge ID.

- If the SW2-SW3 link becomes unidirectional & SW2's BPDU can't reach SW3 & what will happen? (assume devices didn't detect physical issue)
  - SW3 Gi0/1 will become a designated port & start forwarding BPDUs.

- Because SW3's BPDU are inferior to SW2's, SW2 simply ignores SW3's BPDUs.
- SW2 Gi0/1 and SW3 Gi0/1 are both in forwarding state. which means we'll have loop on SW1-SW3-SW2. ~~Loop~~

Loop Guard: The solution.

When a Loop Guard Enabled Port's Max Age timer counts down to 0, it doesn't became a Designated port. and start transitioning to Forwarding.
  - It enters the Broken (Loop Inconsistent) state.
  - Like the Broken (Root Inconsistent) state triggered by a Root Guard violation, this blocks the port.
  - Note: In both cases, the port remains up/up but STP blocks it.

- If the broken port starts receiving BPDUs again, it will be automatically re-enabled.
- Loop guard can be configured in two ways:

- Per Per Port: SW3(config-if)# spanning-tree guard loop
- Default: SW3(config)# spanning-tree loopguard default
  - This enables loop Guard on all ports

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To disable loop Guard on specific ports use.
SW3 (config-if) # spanning-tree guard none
&

NOB:
Loop Guards should be enabled on Root & Non-designated ports (ports that are supposed to receive BPDUs).

BKN = Broken
LOOP-INC = Loop Inconsistent

Loop Guard & Root Guard

- Loop Guard & Root Guard are mutually exclusive.
  - They can't be enabled on the same port at the same time.
- Root Guard is meant to prevent Designated ports from becoming Root ports.
- Loop Guard is meant to prevent Non-Designated ports or Root ports from becoming Designated ports.
- If Loop Guard is configured on a port (span-tree guard loop) & then we configure Root Guard (span-tree guard root), Loop Guard will be disabled on port & vice versa.
  - ~~There more specific configuration interface vs~~
- If loop Guard is enabled by default (spanning tree loopguard default) and then we configure root guard on

a port, the Loop guard will be disabled on port.

- The more specific configuration (int vs global) takes effect.

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